

Tanishk Yadav

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EDUCATION

New York University, Tandon School of Engineering

Brooklyn, NY

Master of Science in Financial Engineering

05/27

- Coursework: Introduction to Derivative Securities, Valuation, Quantitative Methods, Credit Risk and Financial Risk Management, Econometrics and Time Series Analysis, Real Time Risk Management, Algorithmic Trading and High Frequency Finance, Corporate Valuation

SRM University (100% scholarship, GPA 3.8/4.0)

Amaravati, India

B.Tech in Computer Science & Engineering (AI & Machine Learning)

06/25

- Coursework: Single/Multi Variable Calculus, Differential Equations, Linear Algebra, Probability, Statistics, Economics

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, R
- **Libraries & Frameworks:** Pandas, NumPy, SciPy, Statsmodels, Scikit-Learn, TensorFlow/Keras, XGBoost, hmmlearn
- **Financial Concepts:** Options Pricing (Black-Scholes, Greeks), Implied Volatility (Brent root-finding), Volatility Risk Premium, Monte Carlo Simulation, VaR/CVaR, Regime Switching (HMM, GMM), Tail Risk Hedging, Time Series (ARIMA, AR), Mean-Variance Optimization, Walk-Forward Validation
- **Tools & Platforms:** Bloomberg Terminal, Git, LaTeX, Jupyter, Excel, VS Code
- **Certifications:** Bloomberg Market Concepts (BMC), Bloomberg Finance Fundamentals, Akuna Capital Options 101, Financial Markets (HONORS), AI for Investments

EXPERIENCE

IUDX (India Urban Data Exchange), Bengaluru, India

06/23 – 08/23

Summer Intern, Framework Development

- Developed a secure, cross-platform digital consent collector using SHA-256 cryptographic hashing, ensuring regulatory compliance for critical data aggregation on the ADeX platform, a government-academia joint venture
- Engineered full-stack JavaScript solutions for consent collection and retrieval as part of a 4-person team, ensuring 100% compliance with central govt. regulations and enabling creation of 38 critical datasets for the government

PROJECTS

S&P 500 Sector-Level Clustering & Market Structure Analysis | *Python, Keras, Scikit-Learn*

- Classified **66 GICS sub-sectors across 425+ companies** into monopolistic, duopolistic, and oligopolistic market structures using a custom overperformance index with squared-weight allocation across 5 normalized financial metrics (YoY market cap growth, revenue growth, weighted-vs-simple variance, short/long-term beta covariances)
- Trained a multi-layer perceptron neural network achieving **93% classification accuracy** across 4 sector clusters identified via hierarchical and K-Means clustering; published as standalone research with full reproducibility across 13 modular notebooks (ORCID: 0009-0006-2382-9411)

Regime-Aware Fundamental Strategy & Point-in-Time Data Engine | *Python, HMM, SEC EDGAR*

- Architected a production-grade SEC EDGAR pipeline enforcing zero lookahead bias at 4 independent layers (FilingsIndex, FilingSelector, XBRLParser, BloombergMapper), each gated by `acceptance_datetime` assertions for second-level PIT precision; **65 unit tests** across cutoff logic, CIK mapping, schema validation, and Bloomberg parsers
- Built a walk-forward HMM equity strategy across 18 years (2008–2026) delivering **763% total return (633% net of 2/25 fee structure)**, with **8.37% annualized Fama–French 5-factor alpha** ($t = 2.76$, $p = 0.006$), defensive market beta of 0.77, and composite ranking on 4 fundamental factors (ROIC, leverage, earnings stability, FCF yield) smoothed with 4-period EWMA

VolEdge: Volatility & Distribution Analytics Platform | *Python, FastAPI, React, Black-Scholes*

- Built a full-stack platform fitting Gaussian Mixture Models to time-at-price and volume-weighted distributions; computes all Greeks and implied volatility locally via Brent-root Black-Scholes mapped to a 3D IV surface with 25Δ skew, volatility risk premium (VRP), and term-structure diagnostics
- Architected a walk-forward backtest engine with AST-validated sandboxed Python execution, strict $t-1$ data boundaries, and a 3D parameter sensitivity surface for CAGR/Sharpe robustness analysis; five rule-based signal types including VRP compression, skew mispricing, and term-structure inversion

Regime-Adaptive Sector Portfolio Management | *Python, Scikit-Learn, Walk-Forward*

- Designed a multi-scale regime detection system with 4 orthogonal stress detectors (CUSUM sequential analysis, rolling pairwise correlation, market breadth, Fisher skewness) using fuzzy aggregation to avoid binary threshold artifacts
- Achieved **Sharpe 1.08, max drawdown –5.25%** vs. SPY's –41.71% (**87% reduction**), and Calmar 1.20 vs. 0.28 over walk-forward validation (2009–2025); thresholds calibrated only from training data